

A Self-Consistent Finite-Element Schrödinger–Poisson Drift-Diffusion Solver for Quantum-Confinement Effects in Ultra-Thin-Body Nanoscale Devices

Author One*, Author Two†

Department of Electronics and Communication Engineering

Affiliation, City, Country

*email@domain †email@domain

Abstract—As field-effect transistors scale into the few-nanometre regime, the carriers in the thin body occupy quantised subbands and the classical drift-diffusion (DD) picture of a point-like charge continuum breaks down. We present an open-source, two-dimensional finite-element device solver that couples the steady-state drift-diffusion equations, written in the quasi-Fermi-potential variables and solved by a fully-coupled Newton method with an exact analytic Jacobian, to a self-consistent Schrödinger–Poisson (S–P) quantum-confinement correction. The single-band effective-mass Schrödinger equation is solved slice-by-slice across the confinement direction, and the resulting subband density enters the drift-diffusion assembly as a charge-conserving additive quantum potential, so the analytic Jacobian and the quadratic convergence of the inner Newton solve are preserved. Using a silicon ultra-thin-body p – n junction we show that the ground-state subband spacing rises from 10 meV at a 12 nm body to 161 meV at 3 nm, crossing the room-temperature thermal energy $k_B T$ near an 8 nm body—the onset of strong quantum confinement. The correction redistributes carriers across the body by up to $\sim 19\%$ while conserving charge to 1.6×10^{-5} , leaves the ideal-diode terminal current essentially unchanged (within 0.01%, ideality factor ≈ 1.16), and recovers the classical solver exactly when disabled. Metal-oxide-semiconductor gate control (accumulation $\rightarrow$ depletion $\rightarrow$ inversion) is reproduced on the same framework. The solver is validated against analytic semiconductor physics and provides a compact, extensible platform for quantum-corrected TCAD of scaled nanoelectronic devices.

Index Terms—Schrödinger–Poisson, drift-diffusion, finite-element method, quantum confinement, ultra-thin-body, TCAD, nanoelectronics, coupled Newton.

I. INTRODUCTION

THE continued scaling of CMOS technology into ultra-thin-body silicon-on-insulator (UTB-SOI), FinFET and gate-all-around geometries places device dimensions on the order of the electron de Broglie wavelength. In such bodies the transverse motion of carriers is quantised into subbands: the charge peak is set back from the confining walls, the lowest allowed energy is raised, and the electrostatics that determine threshold voltage and inversion-layer capacitance depart from the classical prediction [2], [3]. Technology computer-aided

design (TCAD) for such devices therefore needs a quantum-corrected transport model that remains numerically robust and inexpensive enough for routine use.

The drift-diffusion (DD) system [1] remains the workhorse of device simulation, but on its own it treats carriers as a point-like continuum whose density follows the electrostatic potential instantaneously. A widely used and computationally attractive remedy is to retain the DD transport equations along the device while adding a self-consistent Schrödinger–Poisson (S–P) correction that captures confinement across the thin dimension [3], [5]. This paper presents a compact, open-source implementation of exactly this strategy on an unstructured finite-element mesh, and characterises the resulting quantum corrections for scaled silicon devices.

Contributions. (i) A two-dimensional, finite-element, quasi-Fermi-potential DD solver with a fully-coupled Newton scheme built on the exact analytic 9×9 element Jacobian; (ii) a self-consistent, charge-conserving S–P correction that enters the DD assembly as an additive quantum potential and therefore leaves the inner Jacobian—and its quadratic convergence—unchanged; (iii) a quantitative study of quantum-confinement scaling in silicon ultra-thin-body junctions, and a demonstration of MOS gate control on the same framework; and (iv) validation against analytic semiconductor physics.

II. NUMERICAL FORMULATION

A. Drift-diffusion core

The steady state is described by the three coupled equations for the normalised electrostatic potential $u = \psi/V_T$ and the electron and hole quasi-Fermi potentials $v = \phi_n/V_T$, $w = \phi_p/V_T$, using De Mari intrinsic scaling [4] so that every unknown is $O(1)$. Carrier densities follow algebraically from Boltzmann statistics, $n = n_i e^{u-v}$ and $p = n_i e^{w-u}$. With N_i the linear (P1) triangle shape functions and $C =$

$(N_D - N_A)/n_i$ the scaled net doping, the Galerkin weak forms are

$$R_i^u = \int_{\Omega} (\nabla N_i \cdot \nabla u + N_i(n - p - C)) d\Omega, \quad (1)$$

$$R_i^v = \int_{\Omega} \mu_n n \nabla N_i \cdot \nabla v d\Omega, \quad (2)$$

$$R_i^w = \int_{\Omega} \mu_p p \nabla N_i \cdot \nabla w d\Omega, \quad (3)$$

expressing Poisson's equation and the two zero-divergence continuity equations $\nabla \cdot \mathbf{J}_n = \nabla \cdot \mathbf{J}_p = 0$. Element integrals use 3-point Gauss quadrature on the reference triangle. The global $3N \times 3N$ system is solved by a damped Newton method using the exact analytic element Jacobian, whose block structure is

$$J = \begin{bmatrix} K_{uu} & K_{uv} & K_{uw} \\ K_{vu} & K_{vv} & 0 \\ K_{wu} & 0 & K_{ww} \end{bmatrix}, \quad (4)$$

with, e.g., $(K_{uu})_{ik} = \int \nabla N_i \cdot \nabla N_k d\Omega + \int N_i N_k (e^{u-v} + e^{w-u}) d\Omega$. Per-step damping caps the largest increment so the exponential density terms cannot overflow from a cold start. Optional Shockley–Read–Hall and Auger recombination, per-region heterojunction band offsets (Anderson's rule), Fermi–Dirac statistics, and insulator/gate (MOS) and Schottky contacts are supported but are secondary to the quantum correction studied here.

B. Self-consistent Schrödinger–Poisson correction

Confinement in a thin body is a one-dimensional effect across the thin dimension. As in standard device TCAD, the mesh is cut into slices along the confinement direction and, on each slice, the single-band effective-mass problem is solved by linear finite elements:

$$\left[-\frac{\hbar^2}{2m^*} \frac{d^2}{dy^2} + U(y) \right] \psi_i(y) = E_i \psi_i(y), \quad (5)$$

with $U_n(y) = -k_B T u(y)$ for electrons (and $U_p = +k_B T u$ for holes). The tridiagonal generalised problem $H\psi = EM\psi$ (stiffness plus potential matrix, consistent mass M) yields the lowest `num_states` M -normalised eigenpairs, summed with a thermal (Boltzmann) subband occupation $g_i = e^{-(E_i - E_0)/k_B T}$ into the quantum density shape $\rho(y) = \sum_i g_i |\psi_i(y)|^2$. This shape is renormalised so each slice carries *exactly the same charge* as the classical density it replaces; the correction therefore only redistributes carriers across the confinement direction, keeping Poisson's equation well posed. It enters the DD assembly as an additive quantum potential Λ in the density exponent,

$$n = n_i e^{u-v+\Lambda_n}, \quad \Lambda = \ln \frac{n_{QM}}{n_{cl}}. \quad (6)$$

Because Λ is held fixed within a Newton solve, the analytic Jacobian is unchanged ($\partial n / \partial u = n$, etc.) and quadratic convergence is retained. An outer Schrödinger–Poisson (Gummel) loop recomputes Λ from the updated potential, under-relaxes it, and iterates to a tolerance; with $\Lambda \equiv 0$ the scheme reproduces the classical solver bit-for-bit. This is a

TABLE I
CONFINEMENT SCALING IN THE SILICON ULTRA-THIN BODY

t_b (nm)	$E_1 - E_0$ (meV)	$(E_1 - E_0)/k_B T$	regime
12	10.1	0.39	quasi-classical
8	22.7	0.88	crossover
5	58.0	2.25	confined
3	161.2	6.24	strongly confined

self-consistent predictor–corrector S–P scheme with quasi-equilibrium subbands quantised across one direction and transported classically along the device; it is not a full non-equilibrium (NEGF) quantum-transport model.

III. RESULTS AND DISCUSSION

All results use a silicon p – n junction, 200 nm long along transport ($N_A = N_D = 10^{15} \text{ cm}^{-3}$), with the body thickness t_b across the confinement direction as the scaling variable; material parameters (permittivity, affinity, gap, mobilities, effective masses) are read from an editable material file. The 5 nm reference device has 833 nodes / 1536 triangles; a full bias sweep runs in 9.7 s (classical) and 15.2 s (S–P) on a single core.

A. Validation against analytic physics

The classical core reproduces textbook junction physics: the built-in potential matches $V_{bi} = V_T \ln(N_A N_D / n_i^2)$, the law of mass action $np = n_i^2$ holds at equilibrium to machine precision, the forward I – V is exponential with ideality factor ≈ 1 , and the currents extracted at the two contacts are equal and opposite (conservation). Enabling the S–P correction leaves all equilibrium identities intact because the per-slice renormalisation is charge-conserving.

B. Quantum-confinement scaling

Fig. 1 shows the ground-state subband spacing $E_1 - E_0$ as the body is thinned. It rises steeply—following the expected $\sim t_b^{-2}$ particle-in-a-box trend—from 10.1 meV at $t_b = 12$ nm to 22.7 meV at 8 nm, 58.0 meV at 5 nm, and 161.2 meV at 3 nm (Table I). The spacing crosses the room-temperature thermal energy $k_B T = 25.9$ meV near $t_b \approx 8$ nm: below this thickness a single subband dominates the occupation and quantum confinement is strong, whereas above it many subbands are thermally populated and the body behaves quasi-classically. This crossover is the practical boundary at which a quantum correction becomes mandatory for accurate electrostatics. Fig. 2 shows the computed electron subband ladder for the 5 nm device; the eigenvalues are correctly ordered and widely spaced relative to $k_B T$.

C. Charge-conserving redistribution and terminal current

Because the transverse electrostatic field is weak in a symmetric junction, the classical transverse density profile is essentially flat. The S–P correction modulates it by up to 19% (Fig. 3) while conserving the slice charge to a relative error of 1.6×10^{-5} : the quantum potential Λ_n spans $[-0.10, +0.10]$ for the 5 nm body. This redistribution is precisely the quantum

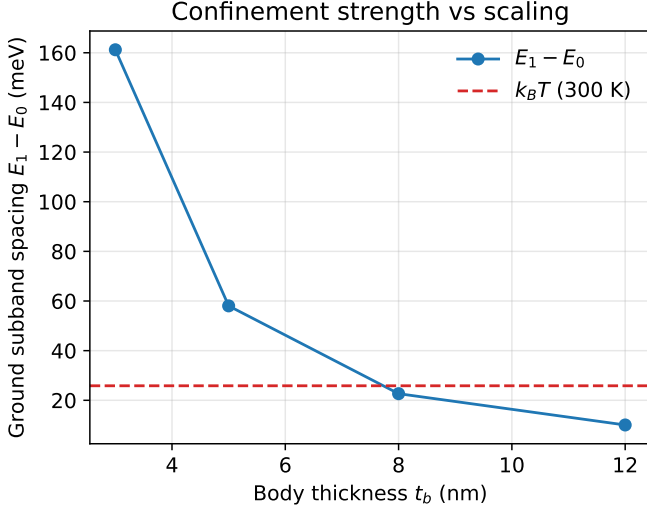


Fig. 1. Ground-state subband spacing $E_1 - E_0$ versus silicon body thickness. Confinement crosses the thermal energy $k_B T$ (300 K) near an 8 nm body, marking the onset of strong quantum confinement.

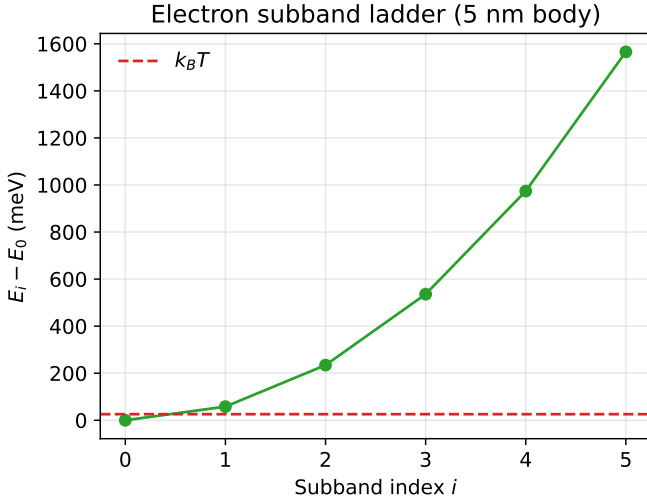


Fig. 2. Electron subband ladder (energies relative to the ground state) for the 5 nm body. The ground-to-first spacing is 58 meV, well above $k_B T$.

electrostatics that shifts inversion-layer capacitance in a gated device. In the ungated diode, however, the terminal I - V is left almost unchanged: the extracted ideality factor is 1.16 both with and without the correction, and the forward current agrees to within 0.01% over 0.4–0.8 V (Fig. 4). This cleanly separates the two roles of the correction—a first-order effect on the charge distribution and capacitance, but a negligible effect on the ideal-diffusion current of an unconfined-transport junction—and confirms that the coupled solve remains rectifying and current-conserving with quantum physics enabled.

D. MOS gate control

The same framework supports insulator (oxide) regions carrying permittivity but no mobile charge, and an insulated

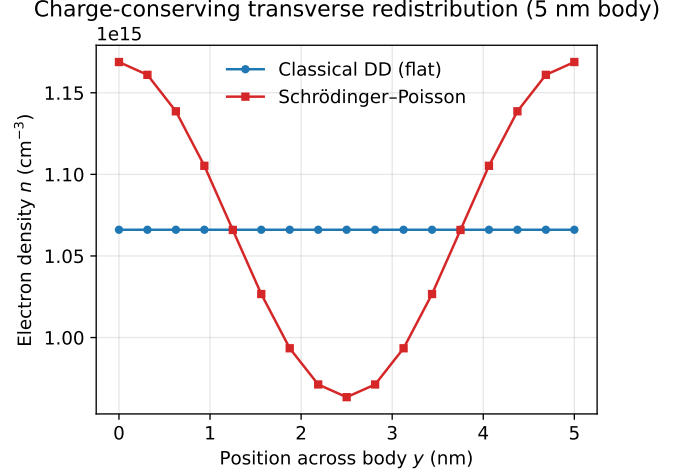


Fig. 3. Electron density across the 5 nm body (n-side slice). The classical profile is flat; the charge-conserving Schrödinger-Poisson correction redistributes carriers transversely by up to 19%.

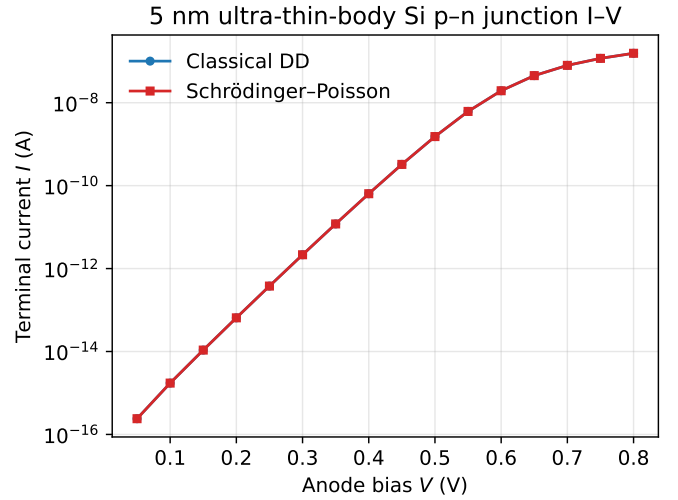


Fig. 4. Forward I - V of the 5 nm ultra-thin-body junction. The quantum correction preserves the ideal-diode current (ideality ≈ 1.16 ; agreement within 0.01%), isolating its effect to the charge distribution.

metal-gate contact referenced through its work function. Fig. 5 sweeps the gate of a silicon/SiO₂ capacitor ($N_A = 10^{17} \text{ cm}^{-3}$, $t_{ox} = 3 \text{ nm}$): the silicon surface passes through accumulation (surface holes $5.2 \times 10^{19} \text{ cm}^{-3}$), depletion, and strong inversion (surface electrons $1.5 \times 10^{20} \text{ cm}^{-3}$), an inversion-to-depletion electron ratio exceeding 10^{11} . This confirms that the solver captures the gate electrostatics of the MOS building block, onto which the quantum inversion-layer correction of the previous sections directly applies.

E. Discussion

The correction is built and normalised per slice, so it is bounded and stable, and is strongest exactly where it should be—thin, strongly-confined bodies—fading as the body thickens and the subband spacing drops below $k_B T$, consistent

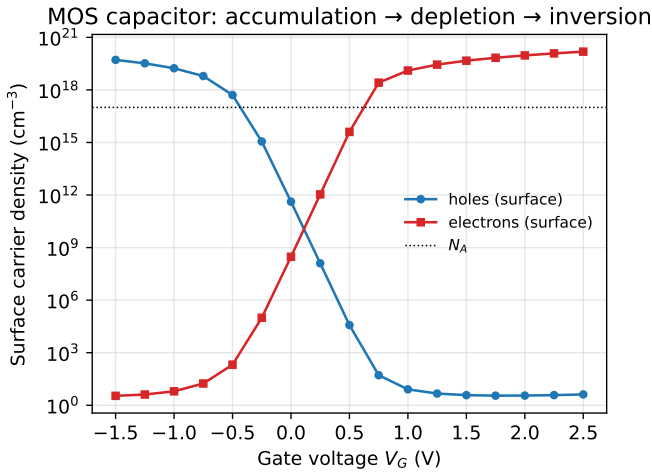


Fig. 5. MOS capacitor surface carrier densities versus gate voltage: accumulation → depletion → inversion, reproduced on the same finite-element framework.

with Fig. 1. Slices are exact on structured/aligned meshes; on a fully unstructured mesh, slices that cannot be formed are left classical. Because the quantum potential enters as a frozen additive shift, the method adds only an outer fixed-point loop around an otherwise unchanged quadratically-convergent Newton solve—a favourable cost trade-off for routine TCAD. The present model is a quasi-equilibrium, single-band scheme; multi-band $k \cdot p$ subbands, thermionic-emission heterointerface transport, and polarisation charge for wide-gap devices are natural extensions.

IV. CONCLUSION

We have presented and validated an open-source, finite-element, self-consistent Schrödinger–Poisson drift-diffusion solver for quantum-confinement effects in scaled nanoelectronic devices. The coupled Newton core retains an exact analytic Jacobian and quadratic convergence, and the charge-conserving quantum correction adds only an outer loop. For silicon ultra-thin bodies the ground subband spacing crosses $k_B T$ near 8 nm, delimiting the strong-confinement regime; the correction redistributes carriers by up to 19% with negligible impact on the ideal-diode current, and MOS gate control is reproduced on the same platform. The tool offers a compact, extensible basis for quantum-corrected simulation of emerging nanoscale devices.

SOFTWARE AND REPRODUCIBILITY

The solver and every example used to generate the figures in this paper are open-source (MIT licence). All results were produced with the bundled structured-mesh device demonstrations.

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